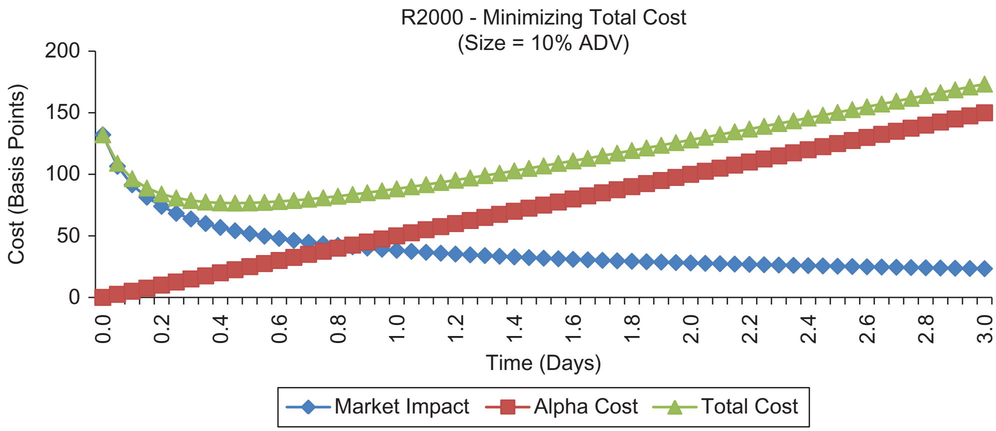
Trade Characteristics	
Size:	10%
Volatility:	43%
Alpha/Day (bp):	100
Alpha/Total (bp):	300

Analysis Results (Basis Points)	
Size:	10%
Volatility:	28%
Min Total Cost:	77
Market Impact:	54
Alpha Cost:	23
Time:	0.45
Alpha 3 Days (bp):	300
Net Profit (bp)	223

Profit Analysis (bp)	
Size	Net Profit
1%	282
5%	250
10%	223
15%	201
20%	182
25%	164
30%	148



■ **Figure 11.9** Alpha Capture Process.

a cost equal to the projected alpha of 3% or 300 bp. This is determined by solving the following optimization:

$$\begin{aligned} &Max \quad Shares \\ &s.t. \quad a_1 \cdot \left(\frac{Shares}{ADV}\right)^{a_2} \cdot \sigma^{a_3} \cdot \alpha = 300 \text{ bp} \end{aligned} \tag{11.26}$$

In this example, the manager could purchase up to 91% of the stock's ADV over 1.5 days at a cost of 300 bp. This is provided in Figure 11.9.

Part 3. How much should the manager invest in the stock is determined by performing economic opportunity cost analysis. The portfolio manager will invest dollars in the stock until the net profit is equal to the expected return of the next most attractive vehicle (economic opportunity cost). In this example, the manager can purchase an order equal to 15% ADV resulting in a net profit of 201 bp. Purchasing any more than 15% ADV